ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech.(CSE) (AI)

Semester: VI

SUBJECT CODE: BCSAI602

COMPILER DESIGN

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(10 Marks)**

Q. 1. Answer the following questions: **(1 × 5= 5Marks)**

(i) Define augmented grammar.

(ii) Mention the types of LR parser.

(iii) Differentiate compiler and interpreter.

(iv) Define the following terms: **a)** Lexeme **b)** Token

(v) What are the differences between the NFA and DFA?

Q. 2. Answer the following questions: **(1 × 5= 5 Marks)**

(i) What is input buffering?

(ii) What are Abstract Syntax trees?

(iii) Define left recursion with example.

(iv) Define left factoring with example.

(v) Explain intermediate code representations?

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Construct an DFA equivalent to the regular expression

(0+1)*(00+11) (0+1)*

OR

(b) Explain the various phases of a compiler in detail. Also write down the output for the following expression**: position: =initial + rate * 60.**

(ii)(a) Construct the predictive parser for the following grammar:

S→ (L) |a

L→L, S |S

OR

(b) Discuss the various types of compilers (e.g., single-pass, multi-pass, cross-compilers). Explain how they differ and provide practical use cases for each type.

Q.4.(i)(a) What is a symbol table in a compiler? Discuss its role in semantic analysis and describe different data structures used for symbol table implementation, comparing their advantages and disadvantages.

OR

(b) What is a syntax-directed definition (SDD)? Explain the difference between S-attributed and L-attributed SDDs with suitable examples.

(ii)(a) Define a Directed Acyclic graph. Construct a DAG and write the sequence of Instructions for the expression: **a + a*(b-c) + (b-c)*d**

OR

(b) What is the purpose of code optimization? Explain in detail loop optimization with example.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20Marks)**

Q.5. (a) Construct an LALR Parsing table for the following grammar:

E→E+T |T

T→T*F|F

F→id

OR

(b). Find the SLR parsing table for the given grammar:

E→E+E | E*E | (E) | id

Q.6. (a) What is a three address code? Mention its types. Translate the following expression**: (a + b) * (c + d) + (a + b + c)** into **a**) Quadruples **b**) Triples.

OR

(b) Explain the structure of an activation record in stack-based allocation. Describe each component of the activation record and its role in procedure execution. Illustrate your answer with a suitable diagram.